

Author Biographies and Contribution Statements

Predicting Student Test Scores with Reliable Score Ranges:
A Leakage-Free Streaming-Rating Study of 2.2 Million Records
from a Private Kazakhstani UNT-Preparation Platform

Applied Sciences, Round 1 Revision

October 6, 2026

This document provides biographies and contribution statements for **Aasso Ziro** and **Ruslan Bulgakov**.

1. Aasso Ziro

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Biography

Aasso Ziro received the Master of Technology degree from Kazakh-British Technical University (KBTU), Almaty, Kazakhstan. He is a Senior Lecturer at the School of Information Technology and Engineering, KBTU, and a Ph.D. candidate at the Faculty of Information Technology, Al-Farabi Kazakh National University (KazNU). His work covers information security auditing, cybersecurity, and the application of machine learning in educational and business settings. He has 7 peer-reviewed publications (9 citations as of 2025), including:

- Ziro, A., Gnatyuk, S., & Toibayeva, S. “Research of the Information Security Audit System in Organizations.” *2023 IEEE International Conference on Smart Information Systems and Technologies (SIST)*, Nur-Sultan, Kazakhstan, 2023.
- Ziro, A., Gnatyuk, S., & Toibayeva, S. “Improved Method for Penetration Testing of Web Applications.” *IntelITSIS 2023 (Intelligent Information Technologies & Systems of International Scholars)*, 2023.

Contributions to This Manuscript

Aasso Ziro validated the domain conventions of the Kazakhstani Unified National Testing (UNT) system—score-scale interpretation, discipline taxonomy, and the pedagogical significance of the platform’s test-type categories. This expertise directly underpinned the pseudo-skill cleaning criteria in Section 3.1 and the per-discipline analysis in Figure 15. He reviewed and confirmed the per-stratum conformal coverage results (Table 13) and the per-discipline R^2 values (Figure 15) against independently maintained evaluation records. He reviewed all manuscript drafts for factual accuracy of the Kazakhstani educational context, verified discipline names and UNT-specific terminology, and contributed to the revision-letter responses for Reviewer 1 Comments 1 and 2.

2. Ruslan Bulgakov

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Biography

Ruslan Bulgakov is a researcher at Al-Farabi Kazakh National University (KazNU), Almaty, Kazakhstan, with expertise in statistical machine learning and uncertainty quantification.

Contributions to This Manuscript

Ruslan Bulgakov contributed expertise in statistical measurement and uncertainty quantification to the design and interpretation of the conformal prediction framework in Section 3.3. He advised on the suitability of the Glicko rating deviation as a Mondrian grouping variable and reviewed the exchangeability assumptions underlying the split-conformal calibration procedure. He independently verified the conformal coverage guarantees in Table 13, cross-checking Wilson confidence interval calculations for all three rating-deviation (RD) bands, and reviewed the component ablation (Table 12) for consistency with the headline accuracy metrics. He provided critical revisions to the Methods (Section 3.3, conformal prediction subsection) and the Discussion (Section 5.3, Score Ranges), and reviewed the revision-letter responses to Reviewer 2 Comments 8 and 10.

Author Change Justification

We respectfully submit this statement in response to the editorial inquiry regarding author changes between the original submission and the Round 1 revision.

The revision adds **Artem Bykov** and **Aadm Ziro** (both at IITU, Almaty) as co-authors. Their names were omitted from the original submission due to an administrative oversight at the time of manuscript system entry. Their contributions are:

- **Artem Bykov** (IITU): Data curation—platform database export and schema documentation; Resources—coordination of data access with the platform operator; Writing—review and editing.
- **Aadm Ziro** (IITU): Data curation—pseudo-skill field identification using platform domain knowledge; Resources; Writing—review and editing.

Neither added author has a competing interest with the manuscript content. All five authors have reviewed and approved the final revised manuscript and agree to be accountable for all aspects of the work. No authors have been removed and the order of existing authors is unchanged.

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